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December 19, 1966

SUPERSEDING
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FEDERAL SPECIFICATION

APRON, LABORATORY

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 This specification covers a bib type laboratory apron, rubber coated on both sides, 45 inches long and 35 inches wide, with grommets and neck and waist tie tapes.

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

CCC-T-191—Textile Test Methods.
PPP-B-636—Box, Fiberboard.

Federal Standards:

Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).
Fed. Std. No. 751—Stitches, Seams, and Stitchings.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standard:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preproduction sample. Unless otherwise specified (see 6.1), before production is commenced, a sample of the laboratory apron to be supplied shall be submitted or made available to the contracting officer or his authorized representative for approval in accordance with 4.2. The approval of the preproduction sample authorizes commencement of production but does not relieve the supplier of responsibility for compliance with all applicable provisions of this specification. The preproduction sample shall be manufactured in the same facilities to be used for the manufacture of the production items.

3.2 Illustration. Figure 1 is furnished for the guidance of the supplier. To the extent of any inconsistencies between the written specifications and the figure, the written specification shall govern.

3.3 Material.

3.3.1 *Base cloth.* The base cloth shall be cotton cloth, plain weave, and shall conform

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to the physical requirements specified in table I when tested as specified in 4.4. Warp and filling yarns shall be singles.

TABLE I. *Basic cloth requirements*

Weight per square yard, ounces (minimum)	Yarns per inch (minimum)		Breaking strength, pounds (minimum)	
	Warp	*Filling*	*Warp*	*Filling*
3.0	56	48	48	30

3.3.2 *Coating compound.* The coating compound shall be butadiene-acrylonitrile copolymers or chloroprene rubber suitably compounded, plasticized with phthalate or phosphate ester plasticizers and pigmented black.

3.3.3 *Coated cloth.* The coating compound shall be applied to both sides of the cloth. Weight of the coating shall be not less than 6.0 and not more than 9.0 ounces per square yard on the outside and not less than 1.0 and not more than 2.5 ounces per square yard on the underside; however, the weights shall be in combination to ensure that the overall weight of the cloth shall be as specified in table II. The coated cloth shall be dusted with the powder specified in 3.3.4. The coated cloth shall be fully vulcanized and shall conform to the physical requirements specified in table II when tested as specified in 4.4.

TABLE II. *Coated cloth requirements*

Characteristic		Requirement	
		Minimum	Maximum
Overall weight, oz./sq. yd.		11.5	14
Breaking strength, pounds:			
Initial	—Warp	64	
	—Filling	45	
After acid resistance test	—Warp	58	
	—Filling	40	
Adhesion of coating, lbs./2 inch width.		4.5	
Resistance to acid.		1	
Resistance to aging.		2	
Low temperature flexibility.		3	
Blocking, scale rating			No. 1

[1] There shall be no appreciable stiffening and brittleness or softness and tackiness and there shall be no sign of acid penetration. Appreciable means a change that is immediately noticeable in comparing the tested specimen with the untested coated cloth.

[2] The specimens shall show no signs of becoming stiff and brittle or soft and tacky.

[3] The coating shall show no cracking or flaking.

3.3.4 *Dusting powder.* The dusting powder for application to the coated cloth shall be whiting, talc, or other finely divided mineral material which does not support mildew growth.

3.3.5 *Tie tapes.* The tie tapes shall be cotton, 1/2 to 7/8 inch wide, and the breaking strength shall be not less than 80 pounds.

3.3.6 *Thread, sewing.* The sewing thread shall be the same as that used in the standard product of the supplier.

3.3.7 *Grommets.* The grommets shall be brass or aluminum, 11/16 ± 1/32 inch outside diameter and 5/16 ± 1/32 inch inside diameter.

3.3.8 *Cement.* When cement is used for adhering the reinforcement patches to the apron body, the cement shall possess good adhesion, flexibility, and all other characteristics compatible with those required of the coating after application to the cloth.

3.4 **Design and construction.** The apron shall be bib type with neck and waist tie tape assembly. The apron shall be made from a single thickness of the coated cloth specified in 3.3.3, with the heavily coated side on the outside of the apron. Dimensions and tolerances shall be as specified in 3.5 when measured as shown in figure 1. There shall be a hem, folded in on the underside, securely stitched around the entire apron, with reinforcement patches for grommets.

3.4.1 *Stitches and stitching.* Stitches and stitching shall conform to Fed. Std. No. 751, and the apron shall be made using stitch type 301 with approximately 8 stitches per inch.

3.4.1.1 *Backstitching.* Thread breaks shall be backstitched not less than 1 inch. Ends of all seams and stitches, when not turned under in a hem or caught in other seams

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or stitching, shall be backstitched not less than 1 inch.

3.5 **Measurements.** Measurements shall conform to the requirements specified in table III. Measurements shall be taken as shown in figure 1.

3.6 **Marking.** The apron shall be marked in a plain and permanent manner with the manufacturer's name or trademark.

TABLE III. *Measurements*

	Inches
A[1]	35
B[1]	45
C[2]	33½
D[2]	11½
E[2]	12
F[2]	11
G[2]	12

[1] Tolerances shall be plus 1 inch, minus 1/2 inch.
[2] Slight deviations from these measurements will be allowed on aprons which are standard products of the manufacturer and where such changes do not lessen serviceability of the apron.

3.7 **Workmanship.** Workmanship shall be first class in every respect and shall conform to the quality of product established by this specification. The aprons shall be free from defects which may affect their durability and appearance.

4. QUALITY ASSURANCE PROVISIONS

4.1 **Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 **Preproduction sample inspection.** When required, examination and testing of the preproduction sample for conformance to all the provisions of this specification shall be made of a completely fabricated rubber coated cloth laboratory apron.

4.3 **Inspection for acceptance.** Except when otherwise specified, inspection shall be in accordance with the provisions set forth in MIL-STD-105.

4.3.1 *Inspection of components.* In accordance with 4.1, the supplier is responsible for insuring that materials and components are manufactured, tested, and inspected to ensure conformance with the requirements of this specification (4.4).

4.3.2 *Visual examination of the end item.* Any defects found during visual examination shall be classified in accordance with the classification of defects listed in table IV. All completed units presented at one time for delivery to the Government shall be considered a lot for the purpose of visual inspection. The unit of product for this examination shall be one completed laboratory apron. For sampling, aprons shall be taken at random from each inspection lot in accordance with MIL-STD-105. The inspection level for each sample unit shall be inspection level II and the acceptable quality levels (AQL's) shall be 2.5 major and 4.0 minor defects.

TABLE IV. *Classification of defects*

Examine	Defect	Classification	
		Major	Minor
Coated cloth	Any hole, cut, abraded area, tear, or rip through material or exposing base cloth	X	
	Any peeling, cracking, sticking, or flaking of coating compound	X	
	Area of excessive coating, e.g., solid runs, ridges, or lumps		X
	Any pit or abraded area not through the material or exposing base cloth .		X

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TABLE IV. *Classification of defects (cont'd)*

Examine	Defect	Classification	
		Major	Minor
Coated cloth (cont'd.)	Any embedded foreign material which upon removal results in a hole or exposed base cloth	X	
	Any embedded foreign matter which upon removal does not expose base cloth		X
	Any tackiness	X	
	Any blister	X	
	Any area of no coating	X	
	Color not black		X
	Color not uniform		X
Construction and workmanship	Any grommet, tie tapes, or reinforcing patch missing or misplaced	X	
	Any reinforcing patch misplaced to a degree that serviceability is seriously affected	X	
	Any reinforcing patch misplaced affecting serviceability but not seriously		X
	Grommets out of alignment by more than 1 inch	X	
	Any grommet improperly clinched to a degree seriously impairing intended use	X	
	Any grommet improperly clinched but can adequately serve its intended purpose		X
	Any repair or patch	X	
Stitching	Not type specified		X
	Needle chews, broken threads, or stitches		X
	Tight tension which may result in thread breaks or loose tension which results in a loosely joined seam		X
	Not backstitched, ends not backstitched		X
	Number of stitches per inch varies from that specified by more than plus or minus 2 stitches		X
Cleanness	Any spot, stain, or foreign matter clearly affecting appearance		X

4.3.2.1 *Finished measurements.* The aprons shall be examined for defects in dimensions. The sample unit and lot size shall be as specified in 4.3.2. Any measurement deviating from the nominal dimensions and tolerances specified in 3.5 shall be classified as a defect. The AQL shall be 4.0 defects per hundred units.

4.3.3 *Inspection of preparation for delivery requirements.* An inspection shall be made to determine that packaging, packing, and marking comply with the requirements in section 5. Defects shall be scored in accordance with table V. For examination of interior packaging the sample unit shall be one shipping container fully prepared for delivery selected at random just prior to the closing operations. Defects of closure listed shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 with an AQL of 4.0 defects per hundred units.

TABLE V. *Classification of preparation for delivery defects*

Examine	Defects
Markings (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.
Materials	Any component missing or damaged.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, or distortion of container.
Contents	Number of packages per container is more or less than required.[1]

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[1] For this defect, one package shall be examined from each shipping container in the sample.

4.4 **Tests.** All components shall conform to the physical and chemical requirements specified herein. Test methods referenced in table VI shall be as specified in CCC-T-191. Any sample tested which deviates from requirements specified herein shall be considered cause for rejection of the entire inspection lot. The Government reserves the right to perform tests on aprons as deemed necessary to determine compliance with the provisions of this specification.

TABLE VI. *Test methods*

Characteristic	Requirement reference	Method
Base cloth:		
Weight of cloth	Table I	5041
Yarns per inch	Table I	5050
Breaking strength	Table I	5100
Coating compound:		
Material identification[1]	3.3.2	
Coated cloth:		
Distribution of coating[1]	3.3.3	
Overall weight	Table II	5041
Breaking strength—		
Initial	Table II	5100
After acid resistance test	Table II	5100 and 4.5.1
Adhesion of coating	Table II	4.5.2
Resistance to acid	Table II	4.5.1
Resistance to aging	Table II	4.5.3
Low temperature flexibility	Table II	4.5.4
Blocking	Table II	5872
Dusting powder:		
Composition[1]	3.3.4	

[1] Certificate of compliance. The contractor shall submit a certificate of compliance to the contracting officer certifying that the requirements for the laboratory aprons have been met as specified in 3.3.2, 3.3.3, and 3.3.4.

4.5 **Test methods.**

4.5.1 *Acid resistance.* A specimen of the coated fabric, approximately 12 inches wide and 24 inches long, shall be so supported as to form a pocket 1 inch deep, 8 inches wide, and 20 inches long with the heavier coated side up to contact the acid. The pocket shall be filled with diluted sulphuric acid (sp. gr. 1.28 to 1.30), and remain filled with the acid for a period of 48 hours. At the end of this period, the specimen shall be thoroughly washed in cold water and dried at room temperature for a minimum of two hours. Five 4 inch wide by 6 inch long specimens shall be cut from the exposed specimen and then tested for conformance to the requirements of table I.

4.5.2 *Coating adhesion.* Coating adhesion shall be determined by method 5970 of CCC-T-191 except as specified in 4.5.2.1, 4.5.2.2, and 4.5.2.3.

4.5.2.1 *Adhesive.* The cement used in preparing test specimens shall be a solventless adhesive (see 6.2).

4.5.2.2 *Preparation of specimen.* After removal from the plates, the specimen shall be cooled in air at standard conditions for a maximum of 24 hours. The coating shall be cut through to the base fabric with a razor or knife to facilitate the initial separation of the coating from the fabric.

4.5.2.3 *Adhesion of specimen.* If the coating is separated from the fabric at an average force greater than that specified in table II, or if the average force necessary to separate the cemented sample at the cement line is greater than that specified in table II, the adhesion of coating requirement shall be considered as passing. If a proper adhesive bond is not obtained with the above method of sample preparation, the coated fabric shall be unacceptable and shall be reported as failing with respect to bonding of coating to coating. The test report shall indicate separation of coating or separation at the adhesive line.

4.5.3 *Resistance to aging.* The aging test shall be conducted in accordance with method 5852 of CCC-T-191 except that the time of exposure shall be 10 days.

4.5.4 *Low temperature flexibility.* A 1 by 4 inch specimen with the long dimension warpwise and a 1 by 4 inch specimen with the long dimension fillingwise shall be exposed for 4 hours at minus 30° ± 2°F. The

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temperature shall be recorded at the lowest point in the chamber. The specimen shall be bent sharply, with the heavier coated side out, over a 1/8 inch diameter mandrel so that the back of the specimen touches within 1/8 inch distance directly behind the rod. The specimen shall be examined visually for evidence of cracking, stiffening, flaking, or separation of coating from the fabric.

5. PREPARATION FOR DELIVERY

5.1 **Packaging.** Packaging shall be level A, B, or C, as specified (6.1).

5.1.1 *Level A.* Aprons, each compactly folded or folded and rolled, shall be packaged in boxes conforming to PPP-B-636, class weather-resistant, V3c, style optional. The boxes shall be waterproof sealed with tape in accordance with the appendix to the box specification.

5.1.2 *Level B.* Twelve aprons, each compactly folded or folded and rolled, shall be packaged in boxes conforming to PPP-B-636, type CF, class domestic. The boxes shall be sealed and closed in accordance with the appendix to the box specification. Alternatively, 12 aprons may be packaged by wrapping in kraft paper and the wrap securely tied with not less than three ties of cotton twine, one of which shall be positioned lengthwise over the top ends and bottom.

5.1.3 *Level C.* The aprons shall be packaged to afford protection against corrosion, deterioration, and damage from the supplier to the initial destination. The supplier may use his commercial practice provided it fulfills the above requirements.

5.2 **Packing.** Packing shall be level A, B, or C, as specified (6.1).

5.2.1 *Level A.* Aprons shall be packed in boxes conforming to PPP-B-636, class weather-resistant, V3c, style optional. The boxes shall be waterproof sealed with tape and reinforced with bands of filament tape in accordance with the appendix to the box specification.

5.2.2 *Level B.* Unless otherwise specified, 48 aprons shall be packed in boxes conforming to PPP-B-636, type CF, class domestic, style optional. The boxes shall be sealed and closed in accordance with the appendix to the box specification.

5.2.3 *Level C.* Aprons shall be packed to insure carrier acceptance and safe delivery to destination in containers conforming to the rules and regulations applicable to the mode of transportation.

5.3 **Marking.**

5.3.1 *Civil agencies.* In addition to markings required by the contract or order, the shipping containers shall be marked in accordance with Fed. Std. No. 123.

6. NOTES

6.1 **Ordering data.** Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(*a*) Title, number, and date of this specification.
(*b*) If preproduction sample required (3.1).
(*c*) Selection of applicable levels of packaging and packing required (5.1 and 5.2).

6.2 **Eastman 910 cement or equal.**

6.3 Federal specifications do not include all types, classes, grades, etc., of the commodities indicated by the titles of specifications, or which are commercially available, but are intended to cover the types, etc., which are suitable to Federal Government requirements.

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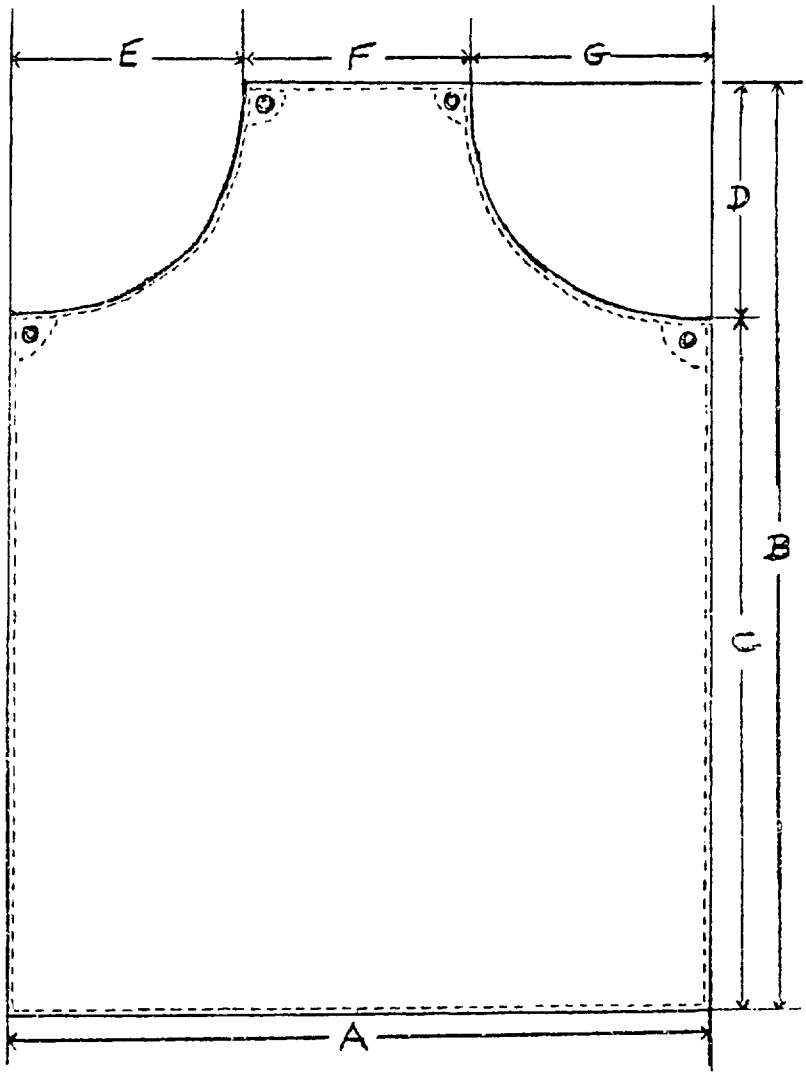


FIGURE 1

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein. Price 5 cents each.

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